ITK-SNAP DLS (nn Intracranial)

Status	Done
Type	

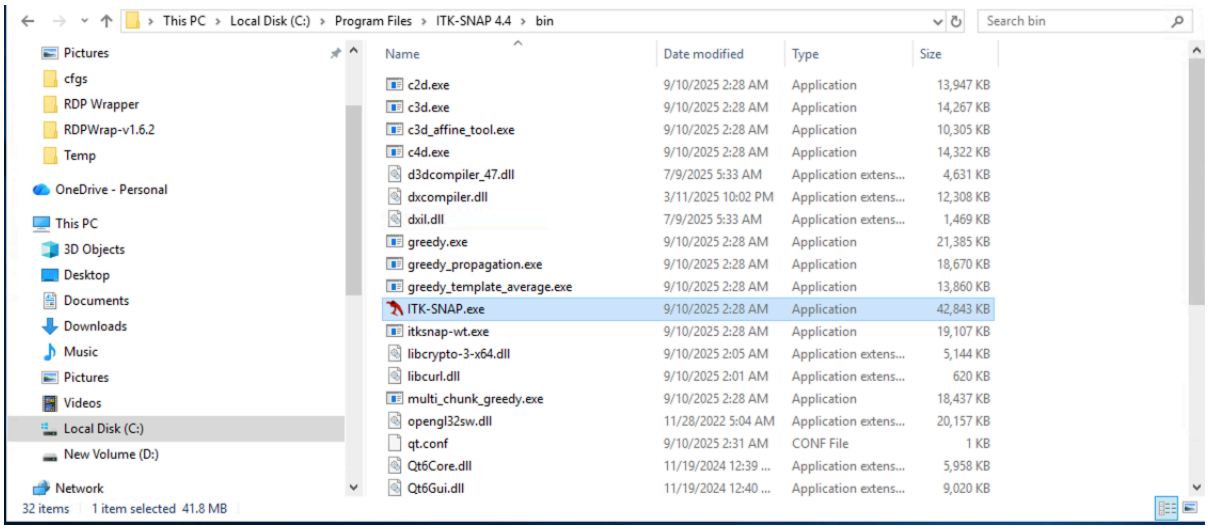
1. Environment Overview
2. Starting the Deep Learning Server (DLS)
    2.1 Open Command Prompt
    2.2 Run the Deep Learning Server
3.Connecting to the Deep Learning Server in ITK-SNAP
    3.1 Open AI extensions settings
    3.2 Deep learning extension server configuration
    3.3 Connect to the Server
4. Loading Label Colors and Descriptions (Required Each Session)
5. Using the AI Segmentation Function

1. Environment Overview

- **Server**: 46
- **GPU computation**: Provided by server 46
- **Required ITK-SNAP version**: **4.4.0 or later**

An appropriate version of ITK-SNAP has already been installed at:

```
C:\Program Files\ITK-SNAP 4.4\bin\ITK-SNAP
```



Please launch ITK-SNAP from this directory.

▼ 2. Starting the Deep Learning Server (DLS)

> **Important**
> At any given time, **only one user needs to start the Deep Learning Server**.
> If the server is already running, other users do **not** need to repeat this step.

▼ 2.1 Open Command Prompt

- In the Windows search bar, type **cmd**



- Open **Command Prompt**

▼ 2.2 Run the Deep Learning Server

In the Command Prompt window, enter the following command:

```
C:\ProgramData\itksnap-dls\venv\Scripts\python.exe -m itksnap_dls
```

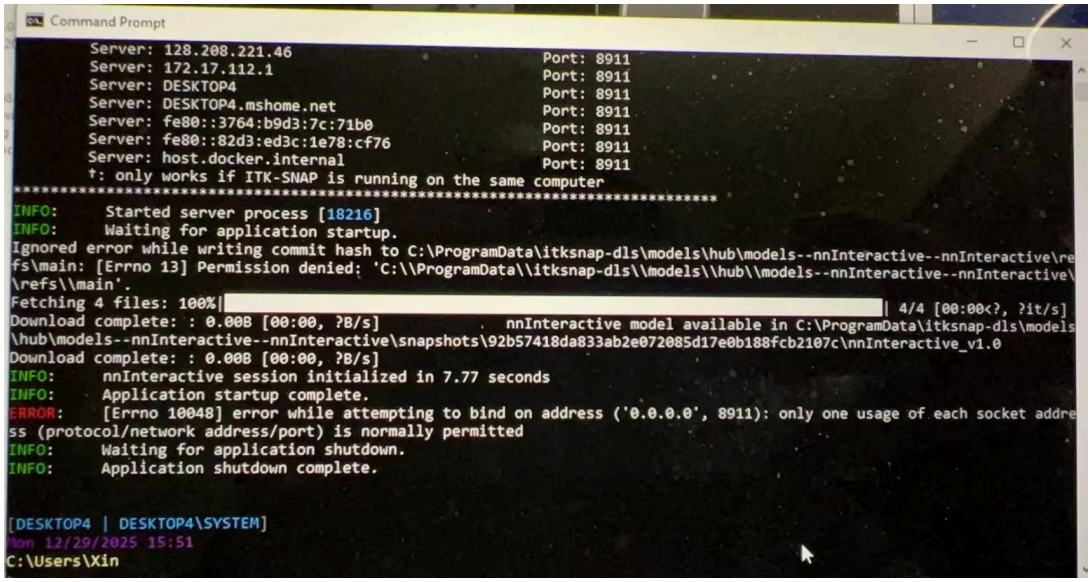
1. **Case A: Server is already running (Expected behavior)**

   If you see an error message similar to:

   ```
   ERROR: [WinError 10048] error while attempting to bind on ..... only on
   e usage of each socket address ...
   ```

   Meaning:

   - The Deep Learning Server has already been started by another user
   - No further action is required
   - You can proceed directly to **Step 3**



2. **Case B: Server starts successfully (Expected behavior)**

   If you see messages similar to:

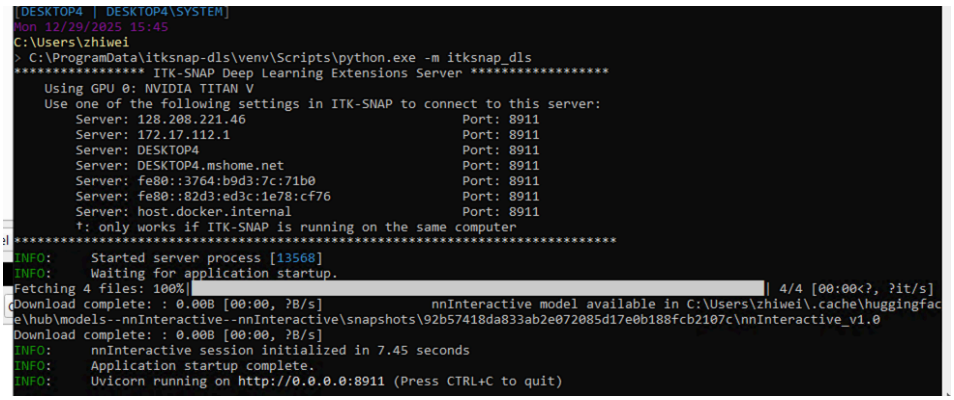
   ```
   Application startup complete
   ```

   Meaning:

   - The Deep Learning Server has started successfully

- GPU-based inference is now available
- You can proceed directly to **Step 3**

> 💡 Do not close this Command Prompt window until you don't need it

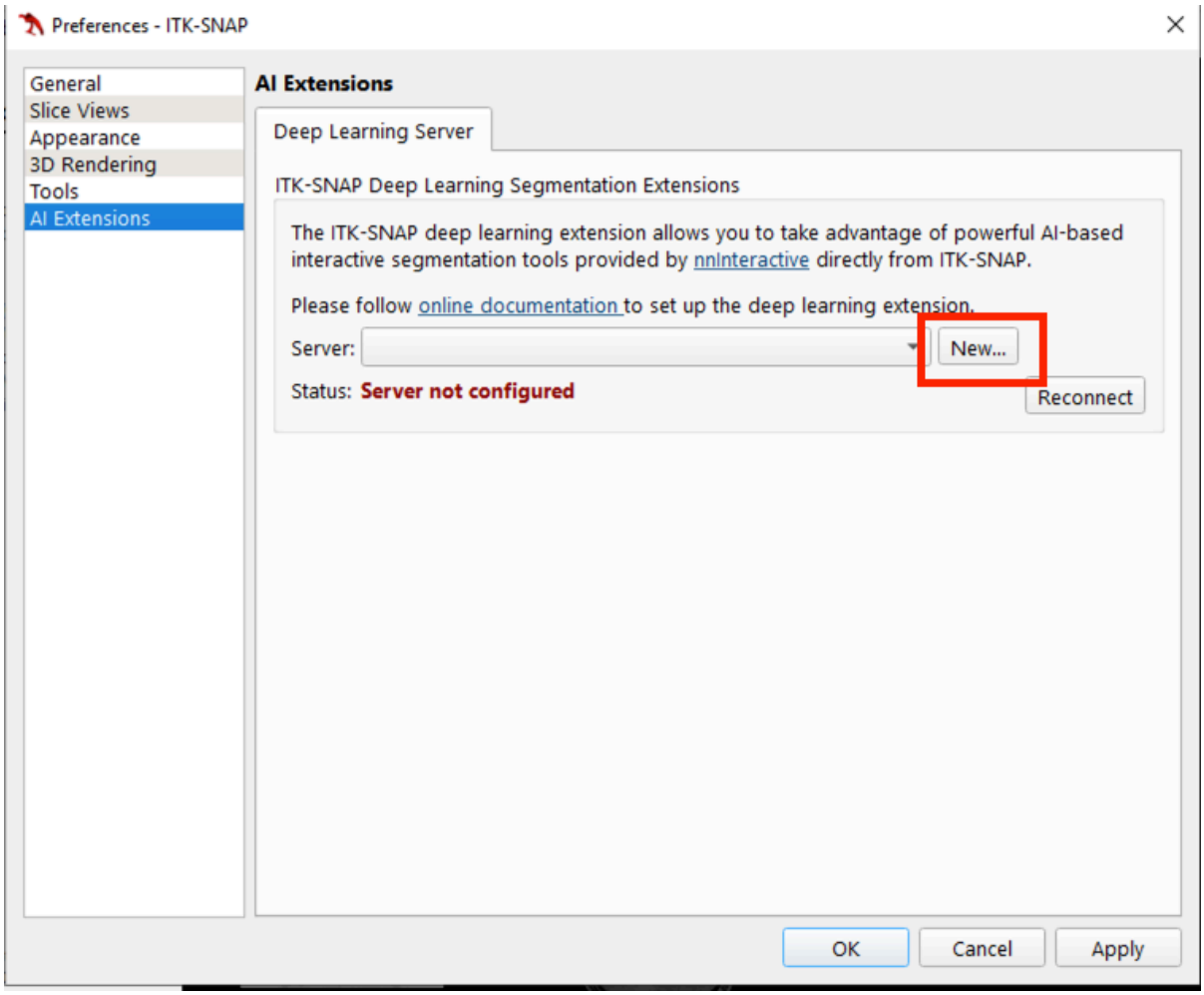


▼ 3.Connecting to the Deep Learning Server in ITK-SNAP

▼ 3.1 Open AI extensions settings

In ITK SNAP, navigate to:

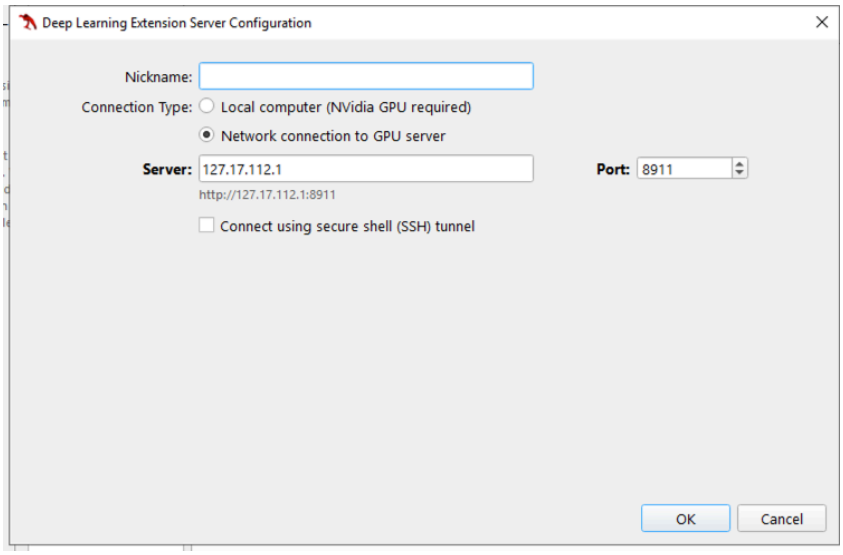
> Tools- preferences - AI Extensions



▼ 3.2 Deep learning extension server configuration

Click **New** or **Edit**, and configure the server as follows:

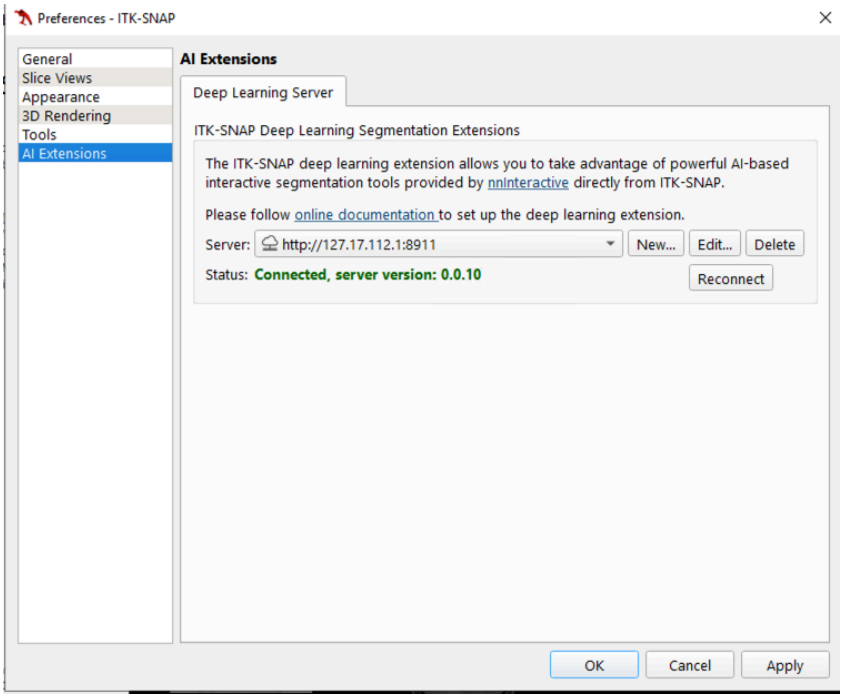
- ☐ network connection to GPU server
- ☐ server: 127.17.11.1
- ☐ Port 8911
- ☐ Click **OK** to return to the AI Extensions panel.



▼ 3.3 Connect to the Server

In the AI Extensions panel:

1. Click **Reconnect**
2. Wait until the status shows connected.



💡 Typically:

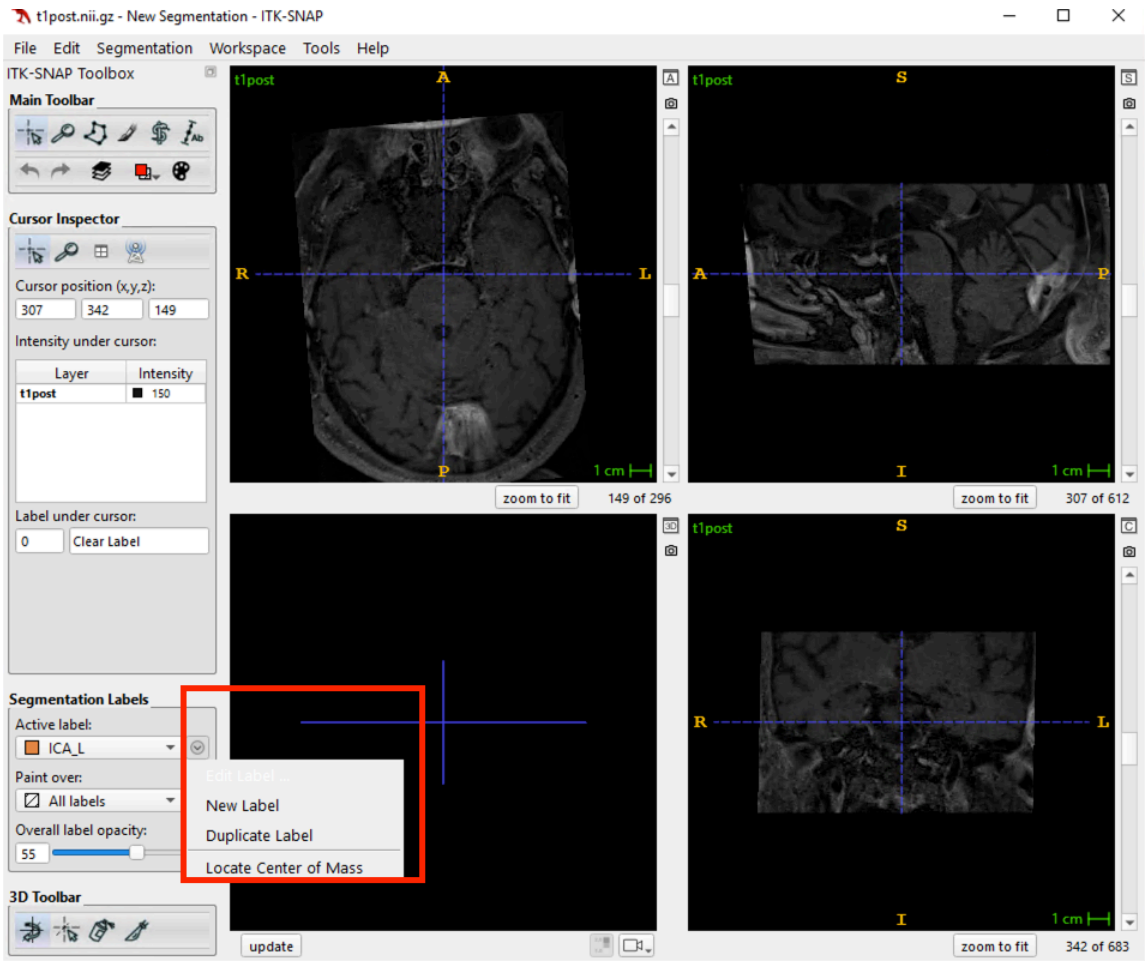
- Full configuration is only required the first time
- On subsequent launches of ITK-SNAP, simply verify that the status is **Connected**

▼ 4. Loading Label Colors and Descriptions (Required Each Session)

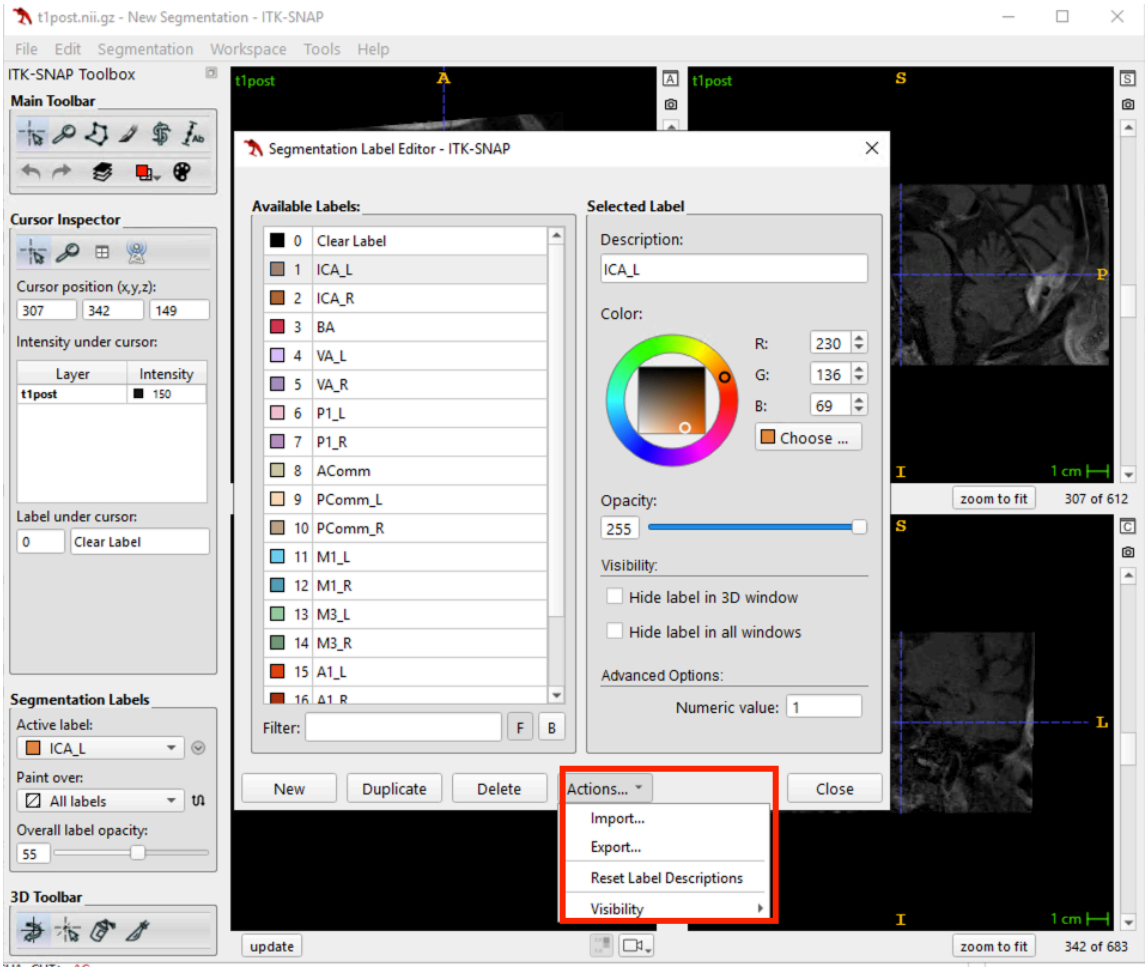
> ⚠️ Important
>
> Before performing AI segmentation, you must load the predefined label colors and descriptions.
>
> This step is required **each time ITK-SNAP is opened**, unless you are working within a saved workspace that already contains the label configuration.

Steps to Load Label Descriptions

1. In the **Paintbrush** panel, locate the **Active Label** section.



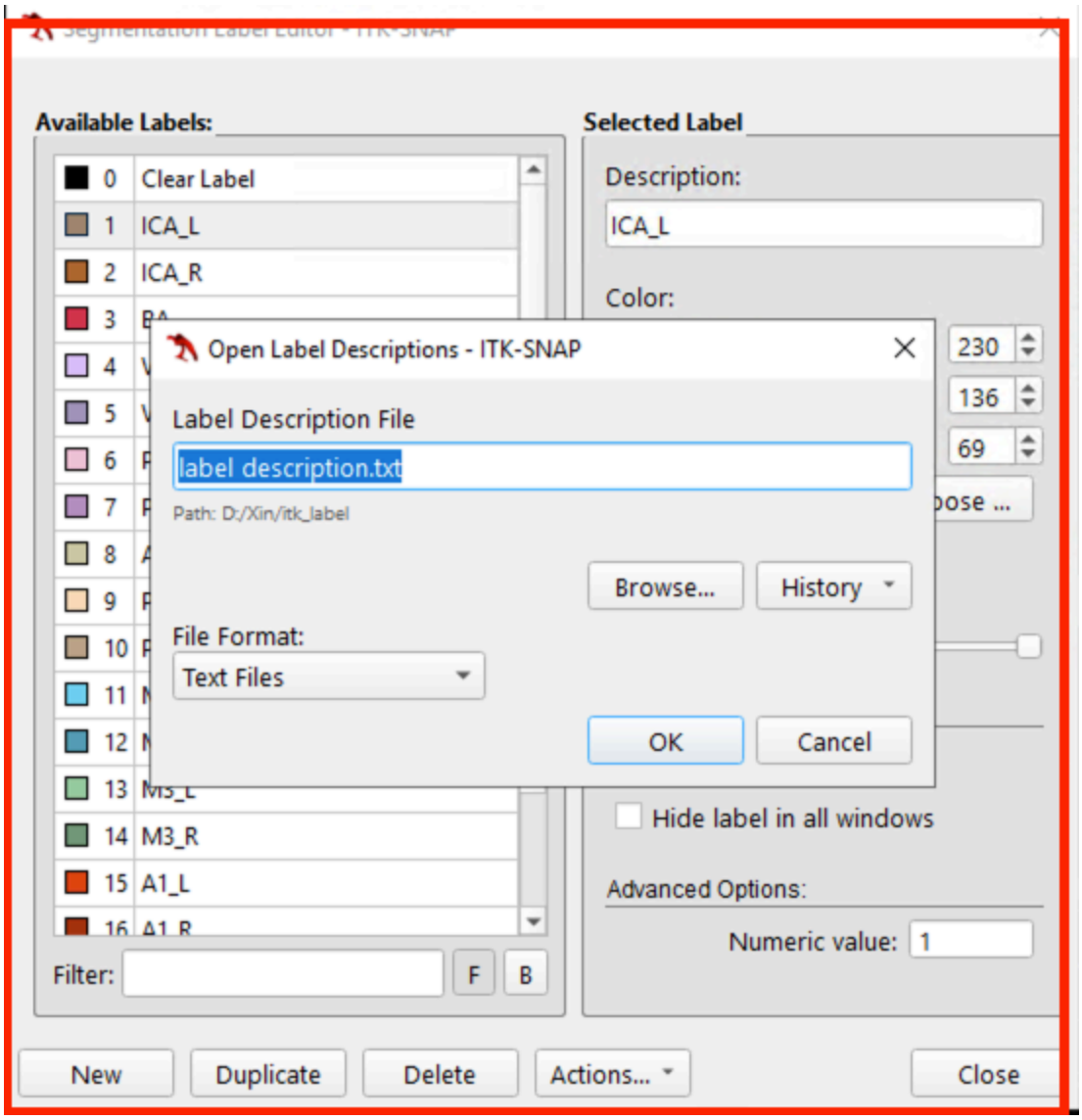
2. Click the **drop-down arrow** to the right of the active label and select **Edit Label**.
3. The **Segmentation Label Editor** window will open.



4. In the **Actions** menu of the Label Editor, select **Import**.
5. In the **Label Description File** dialog, choose the following file:

   D:\Xin\itk_label\label description.txt

6. Click **Open** to import the label definitions.
7. Verify that the label list now shows the correct **label names and colors**.



> 💡 Note
>
> You may copy and save the `label description.txt` file to any location that is convenient for you.
>
> If you do so, make sure to select the correct file path when importing.

▼ 5. Using the AI Segmentation Function

After the label descriptions have been successfully loaded:

1. Return to the main ITK-SNAP window

2. Load the image to be segmented, Ensure the correct label is selected as the **Active Label**.
3. Select the **Paintbrush** tool
4. Enable **AI** mode in the tool options
5. Click on the anatomical structure of interest
6. Wait for the AI-generated segmentation result

